N11: compressed sign-safe Kummer sidecar

25 October 2026 | Proof and derivation packet | No cohort run

State: (K(D), K(D+G), a, n), with D = aG + 2^n T in an odd-prime group.
A branch c in {0,1} sends D to 2D+cG and a to 2a+c modulo r.
The target coefficient is 2^(n+1); induction preserves the invariant.

Sign recovery: if K(X)=K(Y), then Y=X or Y=-X.
If Y=-X and second classes match, then 2X=0 or 2G=0.
The first makes Y=X; the second is impossible for nonzero odd-order G.
Thus (K(X), K(X+G)) is injective, including at identity.
A signed endpoint is checked against aG + (2^n mod r)T.

Across 21 frozen models and 16 jumps: 336 raw rows, 21 group orders.
Explicit pair plus one label: 72 bytes with an external step count.
With a stored counter: 80 bytes. Existing pair plus two labels: 80 bytes.
Regular source path: three xADD calls, 72 field M and 108 field A.
N7 fixed-offset reference: r classes, 32r bytes; different transition.

Mathematical pair-sign ambiguity: zero in the stated group model.
Concrete theta xADD exception bound and signed-lift cost: unknown.
Cohort eligibility: false. No timing or rho-speed claim is made.